RaceDay

Application Documentation — Features Overview

1. Introduction

RaceDay is a full-stack, web-based event management system built for the South African road running, walking, and cycling community. Events such as the Comrades Marathon, the Cape Town Cycle Tour, and hundreds of smaller community park runs and charity cycles are still largely organised through paper forms, spreadsheets, and disconnected communication channels. RaceDay replaces that with a single platform where Event Organisers can create and manage events and Participants can discover events, enter them, and track their own performance history over time.

The system is being delivered in three stages: a planning and database stage, a RESTful API stage, and a web application stage that consumes the API and adds cloud storage and containerisation. This document describes what the finished application will do, from the perspective of the people who will use it.

2. Who Uses RaceDay

RaceDay supports two distinct account types, each with its own registration, login, and permissions.

2.1 Event Organisers

Organisers are the clubs, race committees, and event companies who plan and run races. They use RaceDay to set up events, define categories, monitor who has entered, and capture results once the race is complete.

2.2 Participants

Participants are the runners, walkers, and cyclists who take part. They use RaceDay to browse upcoming events, enter the category that suits them, and look back over their own race history.

Role	Can do
Organiser	Create, edit, and delete their own events
Organiser	Add, edit, and delete race categories within an event (e.g. 5km, 10km, 21km)
Organiser	View everyone enrolled in their events
Organiser	Capture finish times and positions once an event has taken place
Participant	Create an account and manage their own profile
Participant	Browse all upcoming events and their categories
Participant	Enter an event by selecting a category
Participant	View their own current and past enrolments
Participant	Track a personal history of results across every event they've completed

3. Core Features

3.1 Account Registration and Login

Anyone can create a RaceDay account as either an Organiser or a Participant. Once registered, users log in to receive secure, time-limited access to the parts of the system that match their role — Organisers cannot perform Participant-only actions and vice versa.

3.2 Event Management

- **Create events** — Organisers can publish new events with a name, date, location, and description.
- **Edit and delete events** — Event details can be edited at any time before or after the event, and removed if an event is cancelled.
- **Public event browsing** — Anyone — including visitors who haven't logged in — can browse the full list of upcoming events and open one for full details.

3.3 Category Management

- **Multiple categories per event** — Each event can offer multiple categories (e.g. a 5km fun run and a 21km half marathon on the same day), each with its own distance, entry fee, and participant cap.
- **Full lifecycle control** — Organisers keep full control to add, adjust, or remove categories as entries change.

3.4 Event Entry (Enrolments)

- **Enter an event** — A logged-in Participant selects an event and a category to enter, creating a confirmed enrolment.
- **Manage own enrolments** — Participants can see every event they've entered, and cancel an enrolment if their plans change.
- **Organiser visibility** — Organisers can see the full list of everyone entered into their event, broken down by category.
- **Duplicate-entry protection** — The system prevents a participant from entering the same category twice by accident.

3.5 Results Tracking

- **Result capture** — After an event, the Organiser captures each participant's finish time and finishing position.
- **Personal history** — Every Participant can look back through their own finish times and placings across every event and category they've ever completed — a personal performance history.

3.6 Race-Day Preparation

Participants will be able to prepare for race day using live weather conditions for the event location, so they know what to expect and how to dress, alongside the event's location details.

4. Technical Features

Beyond what the user sees, RaceDay is being built with the following underlying capabilities:

- **Relational database** — A relational SQL Server database models Organisers, Participants, Events, Categories, Enrolments, and Results, with primary/foreign keys and constraints enforcing data integrity.
- **RESTful API** — A RESTful API, built in C#, exposes every feature above as a set of secure HTTP endpoints, with role-based access control enforced at the API level.
- **Automated testing & CI/CD** — A unit-tested codebase with an automated GitHub Actions CI/CD pipeline validates the API on every change.
- **Web application (MVC)** — An MVC web application consumes the API to provide the full browser-based experience for both Organisers and Participants.
- **Cloud file storage** — Images and documents (such as event banners) are stored using Azure Blob Storage rather than on the local server.
- **Containerisation** — The finished application is containerised with Docker, making it straightforward to deploy consistently to any environment.

5. Summary

In short, RaceDay gives South African race organisers a single, modern home for running their events — from setup, through entries, to results — while giving participants an easy way to find events, enter them, and keep a lasting record of their own race-day achievements.